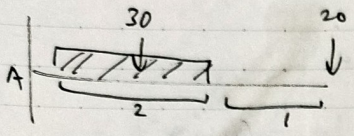
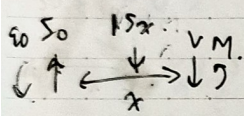
# Question 1

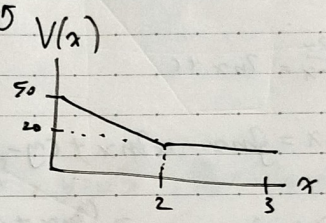



$R_A = 50$ kN.

$M_A = 30 \times 1 + 20 \times 3$

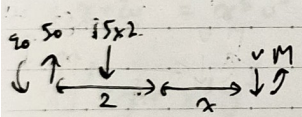
$= 90$ ↻ → Reaction ↺

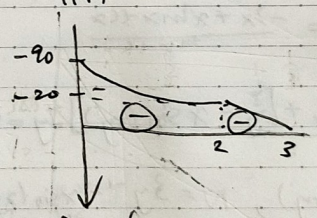



$V = 50 - 15x$.

$-M = 90 - 50x + \frac{15}{2}x^2$

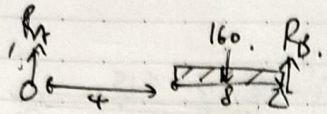



$V = 50 - 30$

$-M = 90 - 50(2+x) + 30 \times (1+x)$ from 2.

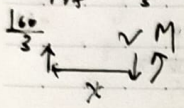
# Question 2



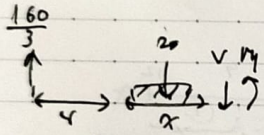

$8 \times 160 = 12 \times R_B$

$R_B = \frac{320}{3}$

$R_A = \frac{160}{3}$




$V = \frac{160}{3}$




$V = \frac{160}{3} - 20x$

$M_0 = 0 = M_{12}$

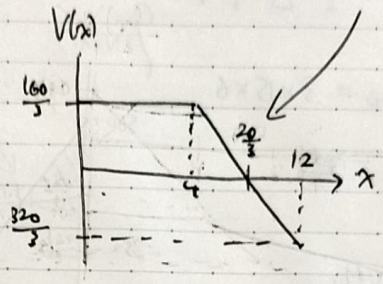
$M_4 = 4 \times R_A = \frac{640}{3}$

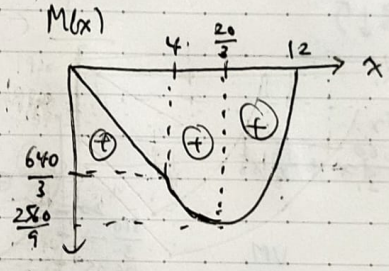
$-M = \frac{20}{2}x^2 - \frac{160}{3}(4+x)$ from 4.

$-M_{\frac{20}{3}} = \frac{20}{2}\left(\frac{8}{3}\right)^2 - \frac{160}{3}\left(4+\frac{8}{3}\right)$

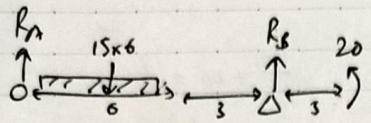
$= \frac{2560}{9}$

$\frac{160}{3} \div 20 = \frac{8}{3}$

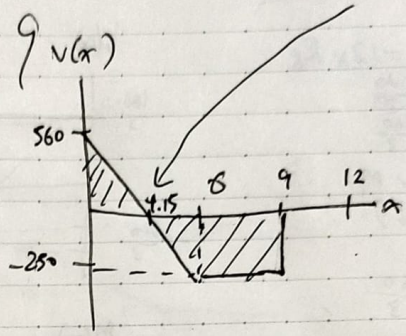


# Question 3




$\frac{560}{9} \div 15 = 4.15$

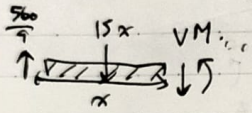



$R_B \times 9 + 20 = 3 \times 15 \times 6$

$R_B = \frac{250}{9}$

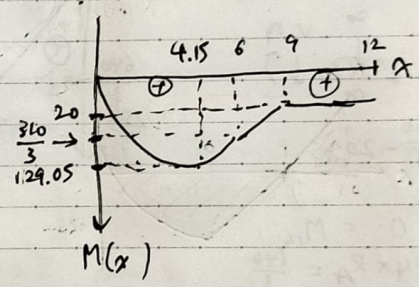
$R_A = 15 \times 6 - \frac{250}{9}$

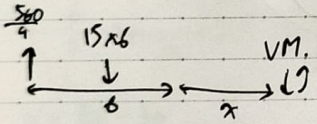
$= \frac{560}{9}$




$V = \frac{560}{9} - 15x.$

$-M = -\frac{560}{9}x + \frac{15}{2}x^2$



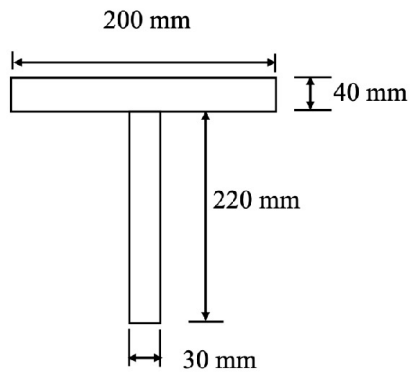

$V = \frac{560}{9} - 15 \times 6$

$-M = -\frac{560}{9}(6+x) + 15 \times 6 \times (3+x)$

$V = 0 \therefore \frac{dM}{dx} = 0 \therefore M = 20$ for $x \geq 9$.

# Question 4

**Q4.** Calculate the second moment of area of the following T-section about the centroidal axes (15 marks).




Let top rectangle be 1, bottom rectangle be 2. Using centroid distance from top left corner.

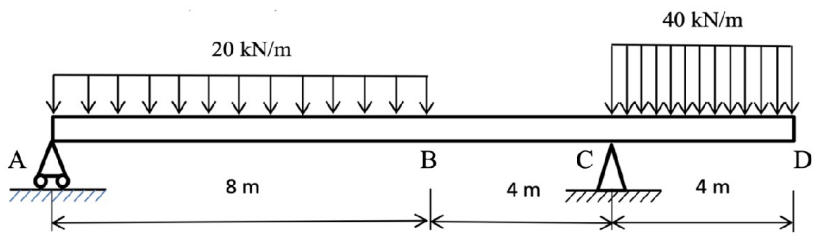
$$\begin{aligned}
\bar{x} &= 100\ mm \\
A_1 &= 200 \times 40 = 8000\ mm^2 \\
\bar{y_1} &= 20\ mm \\
A_2 &= 220 \times 30 = 6600\ mm^2 \\
\bar{y_2} &= 40 + 110 = 150\ mm \\
\bar{y} &= \frac{8000 \times 20 + 6600 \times 150}{8000 + 6600} \\
&= 78.77\ mm
\end{aligned}$$

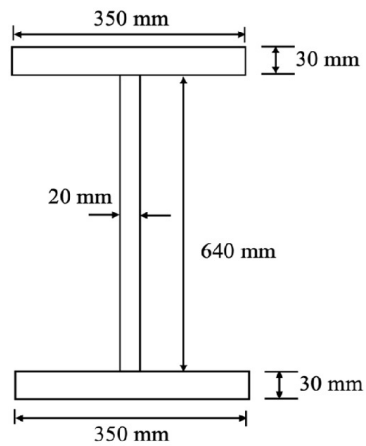
$$I_{xi} = \frac{bh^3}{12}, \quad I_{yi} = \frac{hb^3}{12}$$

$$\begin{aligned}
I_x &= I_{x1\to\bar{y}} + I_{x2\to\bar{y}} \\
&= \frac{b_1 h_1^3}{12} + A_1(\bar{y_1} - \bar{y})^2 + \frac{b_2 h_2^3}{12} + A_2(\bar{y_2} - \bar{y})^2 \\
&= \frac{200 \times 40^3}{12} + 8000 \times (20 - 78.77)^2 + \frac{30 \times 220^3}{12} + 6600 \times (150 - 78.77)^2 \\
&= 88.80 \times 10^6\ mm^4 \\
I_y &= I_{y1\to\bar{x}} + I_{y2\to\bar{x}} \\
&= \frac{h_1 b_1^3}{12} + \frac{h_2 b_2^3}{12} \\
&= \frac{40 \times 200^3}{12} + \frac{220 \times 30^3}{12} \\
&= 27.16 \times 10^6\ mm^4
\end{aligned}$$

# Question 5

**Q5.** For the following beam, calculate the maximum positive (sagging) and negative (hogging) bending moments. Hence, determine the maximum tensile and compressive stresses at the cross sections of maximum positive and negative moments, respectively. (40 marks).




Let top rectangle be 1, middle rectangle 2. Using centroid distance from top left corner.

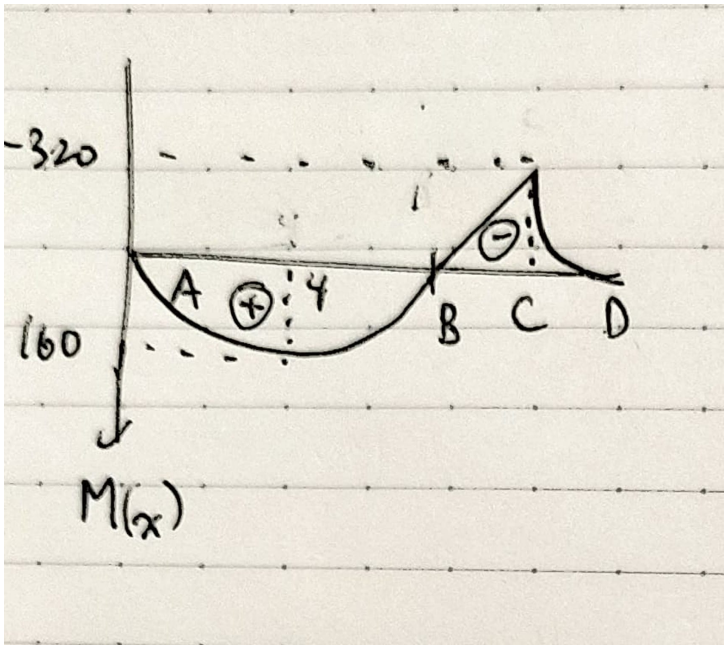
$$\bar{y} = 30 + \frac{640}{2} = 350\ mm$$

$$A_1 = 350 \times 30 = 10500\ mm^2 = A_3$$

$$A_2 = 640 \times 20 = 12800\ mm^2$$

$$\begin{aligned} I_x &= I_{x1\to\bar{y}} + I_{x2\to\bar{y}} + I_{x3\to\bar{y}} \\ &= 2(\frac{350 \times 30^3}{12} + 10500 \times (350-15)^2) + \frac{20 \times 640^3}{12} \\ &= 2.80 \times 10^9\ mm^4 \\ &= 2.80 \times 10^{-3}\ m^4 \end{aligned}$$

$$\begin{aligned} \sum A &= 0 \\ 20 \cdot 8 \times 4 + 40 \cdot 4 \times (8+4+2) &= R_C \times (8+4) \\ R_C &= 240\ kN \\ R_A &= 20 \cdot 8 + 40 \cdot 4 - 240 \\ &= 80\ kN \\ -M_{x:0\to 8} &= 10x^2 - 80x \\ -M_{>8,\ x:0\to 4} &= 20 \cdot 8 \times (4+x) - 80(8+x) \\ -M_{>12,\ x:0\to 4} &= 20 \cdot 8 \times (8+x) + 20x^2 - 240x - 80(12+x) \end{aligned}$$




Taking tensile stress positive, $y$ downwards positive. Maximum positive moment and respective values:

$$\begin{aligned}
M_C &= -20 \cdot 8 \times (4+4) + 80(8+4) \\
&= -320\ Nm \text{ (hogging)} \\
\sigma^+_{max} &= \frac{M_D \times y_{max}}{I_x} \\
&= \frac{-320 \times 350}{2.80 \times 10^{-3}} \\
&= -40\ MPa \\
\sigma^-_{max} &= 40\ MPa
\end{aligned}$$

Maximum positive moment and respective values:

$$\begin{aligned}
M_4 &= -10 \cdot 4^2 + 80 \cdot 4 \\
&= 160 \text{ (sagging)} \\
\sigma^+_{max} &= \frac{M_4 \times y_{max}}{I_x} \\
&= \frac{160 \times 350}{2.80 \times 10^{-3}} \\
&= 20\ MPa \\
\sigma^-_{max} &= -20\ MPa
\end{aligned}$$

Note that $\sigma^+$ denotes the bottom of the beam, and $\sigma^-$ denotes the top of the beam.